CHAPTER Four

Structural Geology

PRIMARY AND SECONDARY STRUCTURES

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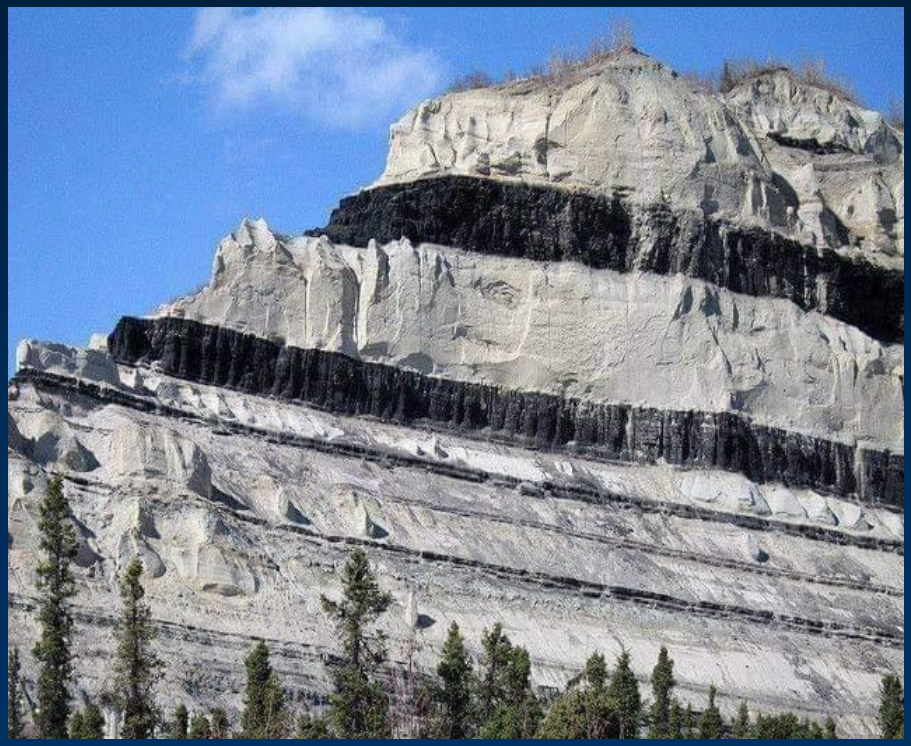
PRIMARY SEDIMENTARY STRUCTURES

التراكيب الاولية

- Bedding or strata طبقات

- Distinct layers having variations in texture, color or physical properties

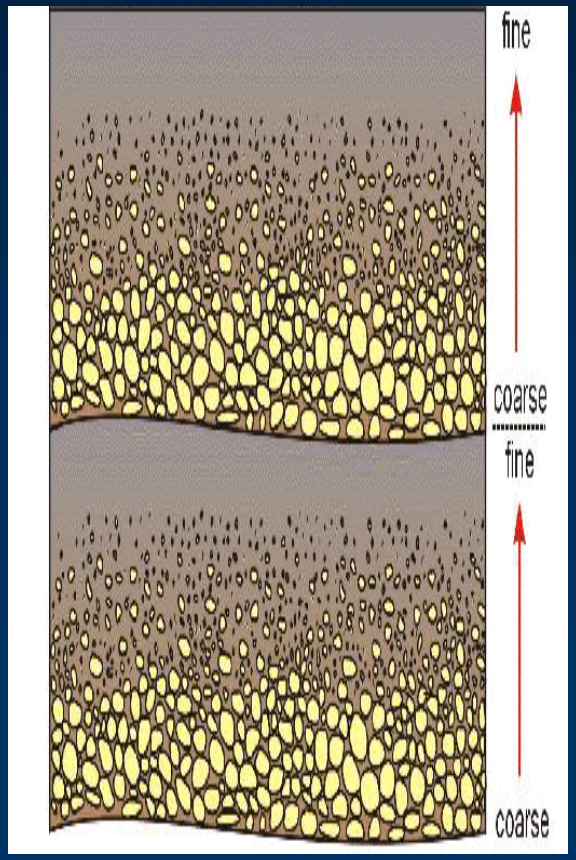


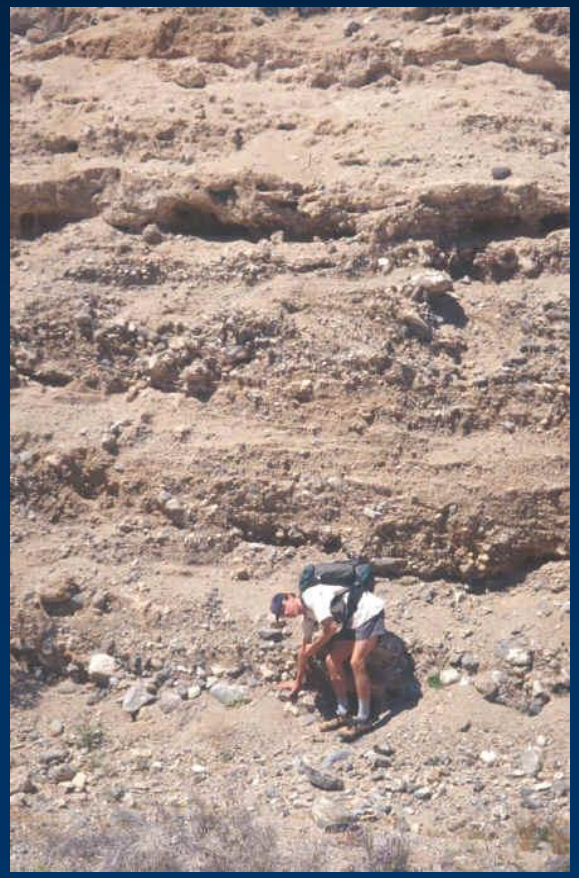
. **Bedding planes**: The planes separating these beds

. Each bedding plane marks the end of one episode of sedimentation and the beginning of another

Graded bedding التطابق المتدرج

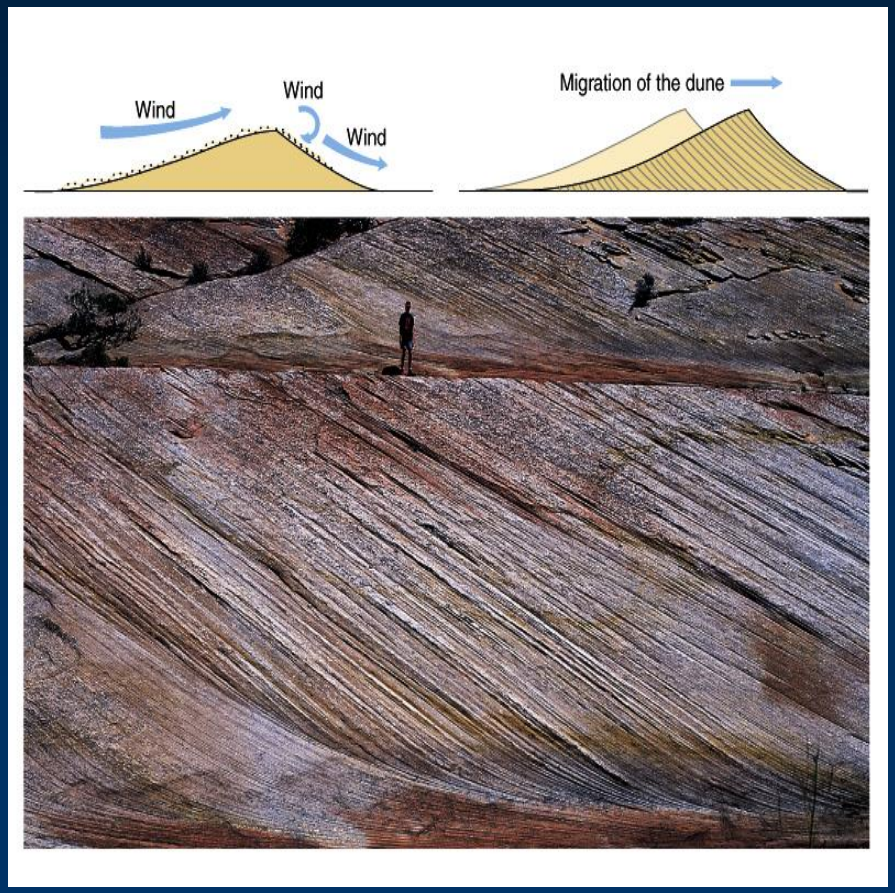
- Progressive change in grain size upward (Fining upward)
- It is characteristic of rapid deposition from water energy decreases, coarse grains deposit first, and in time smaller particles settle to produce graded bedding.



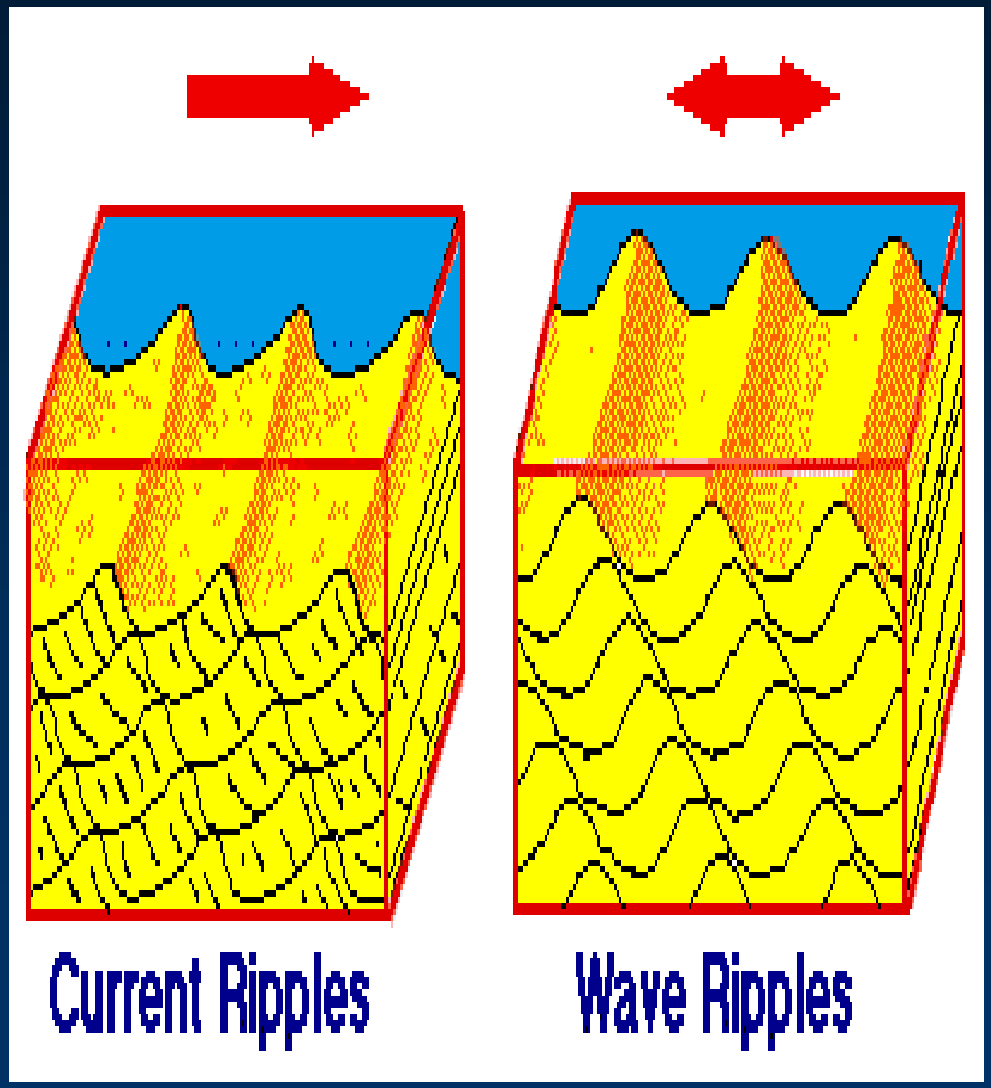


• Cross-bedding التطابق المتقاطع

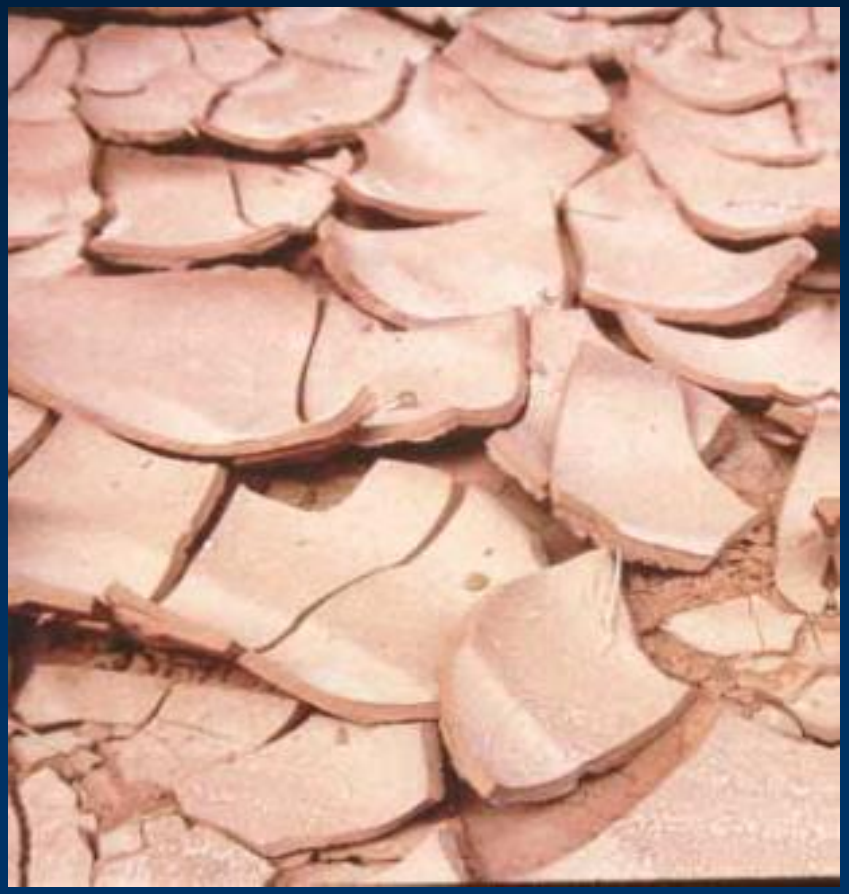
- Individual layers are inclined in reference to the top or bottom of the larger bed
  - Formed by movement of sand waves or dunes
  - Inclination shows flow direction and right-side-up



Ripple marks علامات التماوج



Mudracks التشققات الطينية

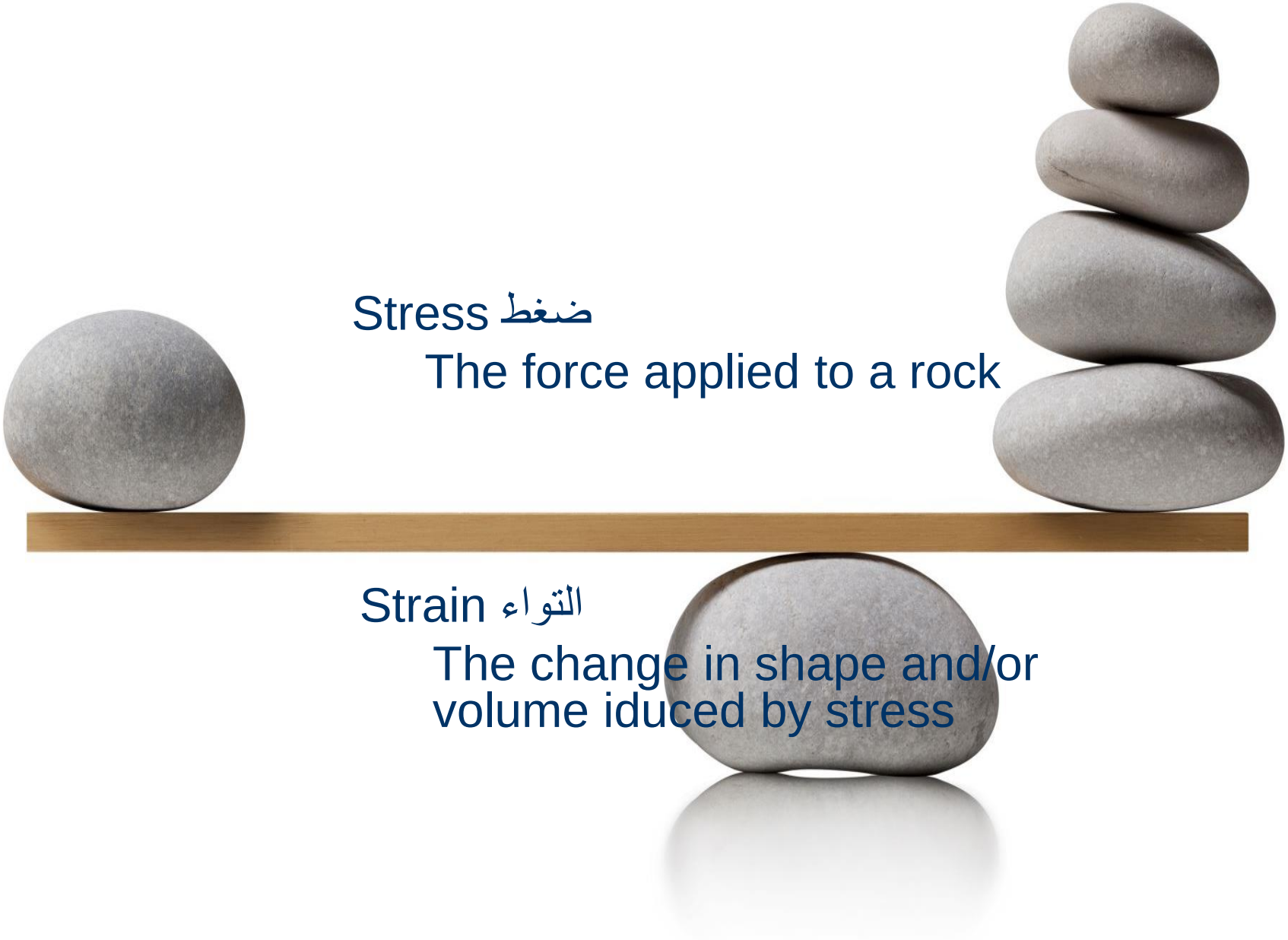


-For interpreting the depositional environment in terms of processes, water depth, wind strength etc.

-For deducing the palaeocurrent pattern and palaeogeography.

SECONDARY STRUCTURES

التراكيب الثانوية

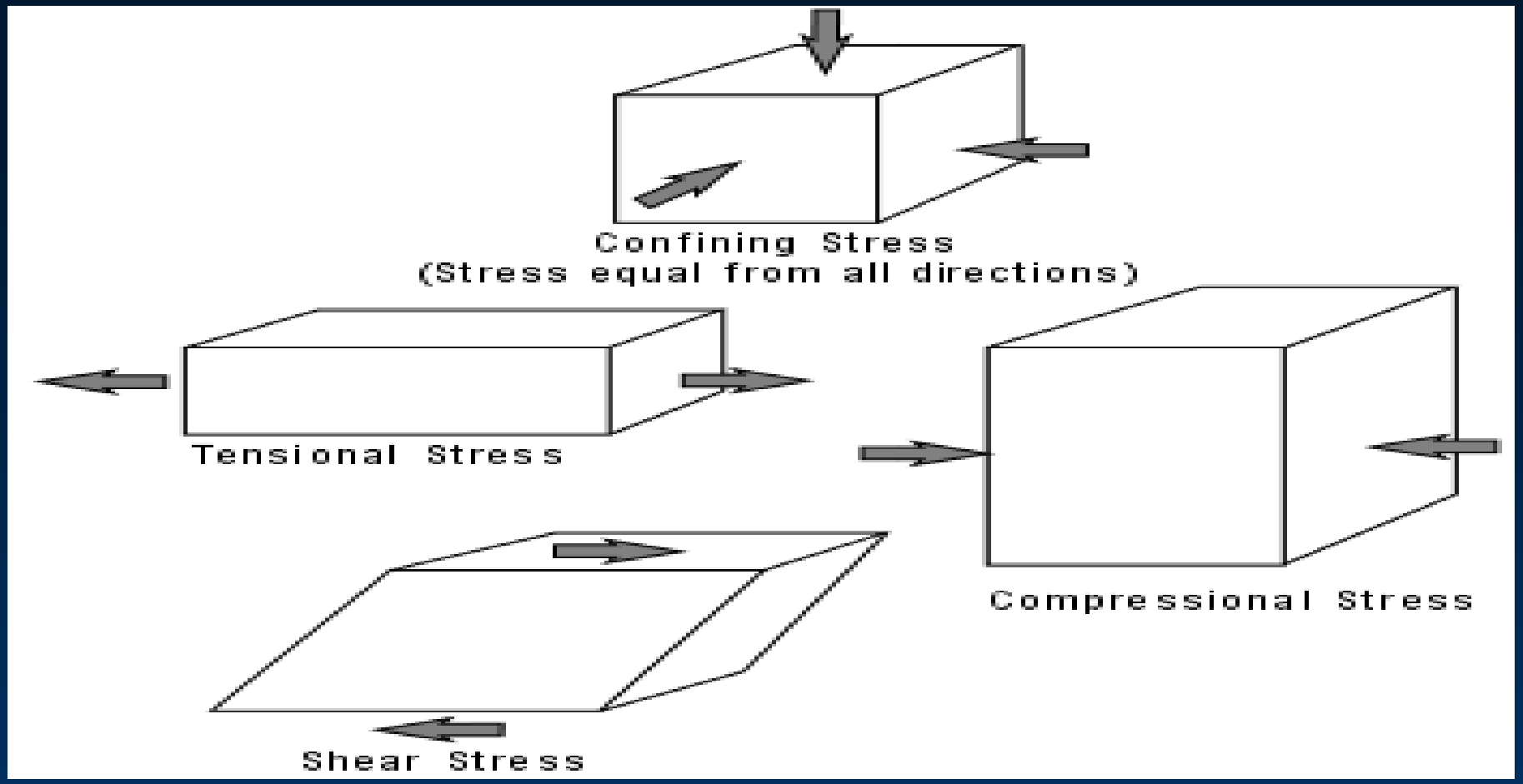


Stress ضغط

The force applied to a rock

Strain التواء

The change in shape and/or volume iduced by stress



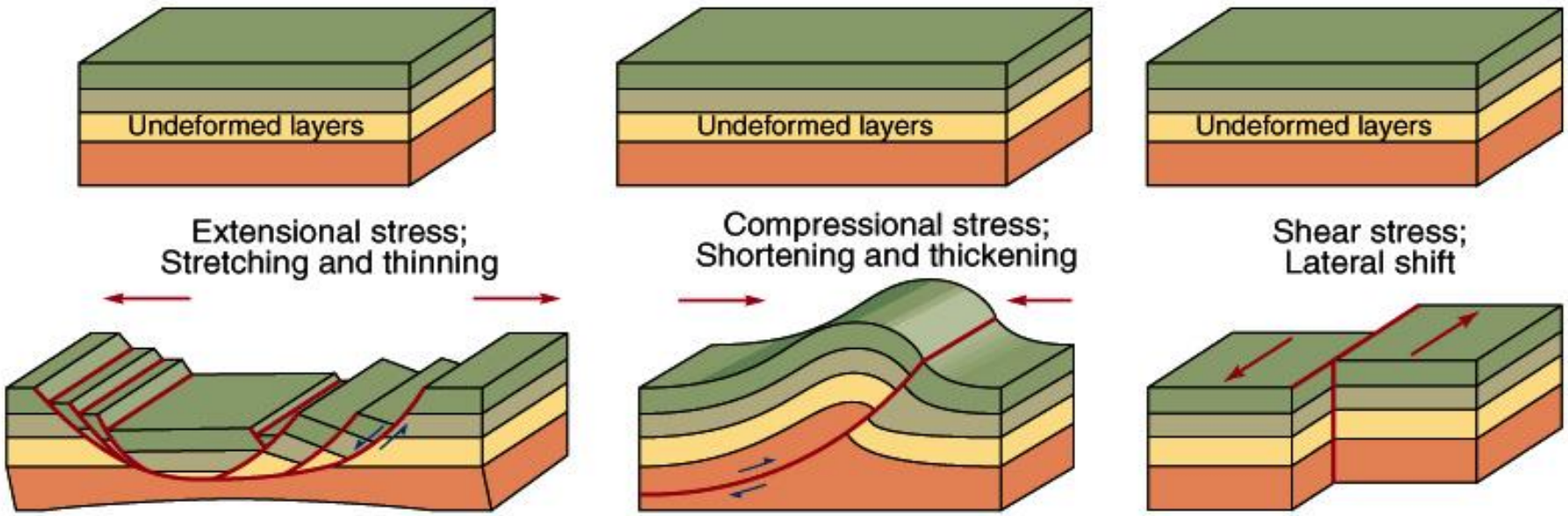
Three types of differential stress

Tensional - شد pulling apart سحب بعيدا

Compressional - ضغط squeezing together

Shear -القص slipping, twisting الانزلاق، التواء

- **Stages of Deformation**
- When a rock is subjected to increasing stress it passes through 3 successive stages of deformation.
- 1- **Elastic Deformation** مرن-the rock retains its back to its original size and shape when the stress is removed.
- 2- **Plastic Deformation** –the rock putty and retains its new shape. معجون ويحتفظ شكله الجديد
- 3- **Fracture** - **t**he rock deformed plastically and then breaks**.**

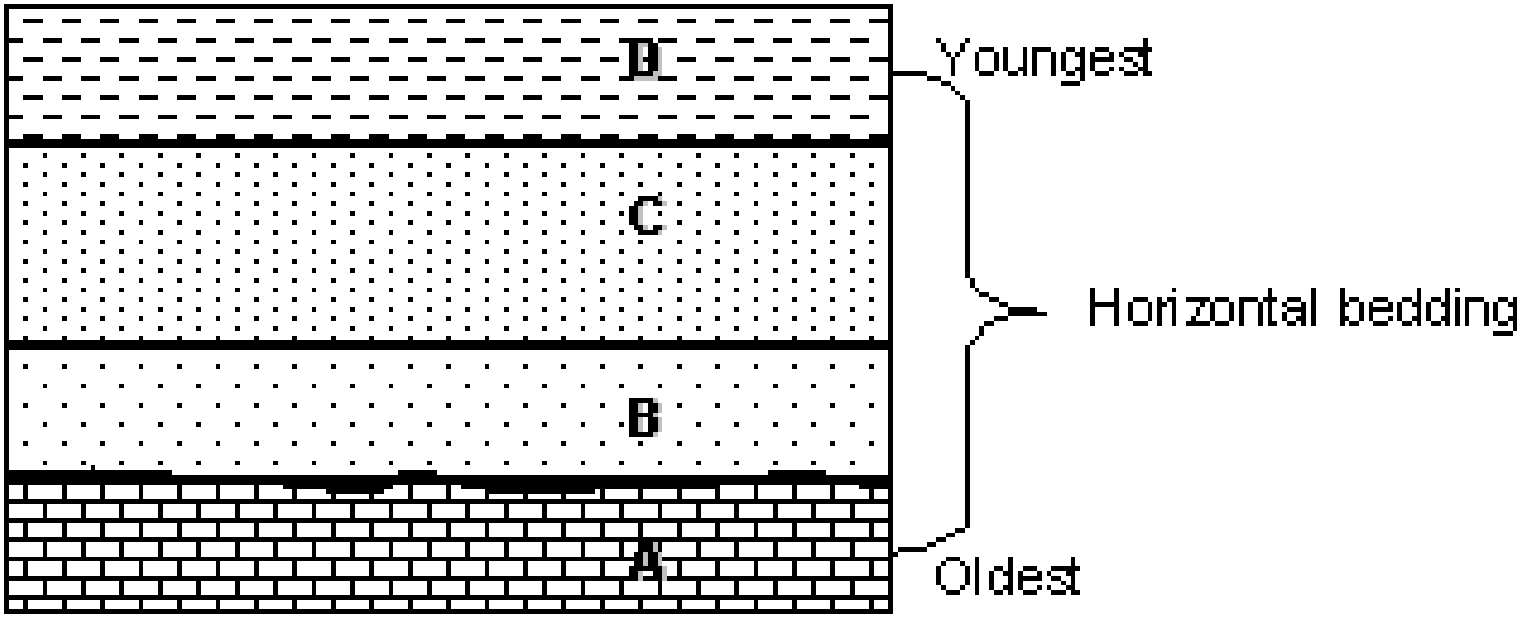

Undeformed layers
Undeformed layers
Undeformed layers
Extensional stress;
Stretching and thinning
Compressional stress;
Shortening and thickening
Shear stress;
Lateral shift

Folds الطيات

- Warps in rock strata due to ductile deformation
  - Generally indicate horizontal compression

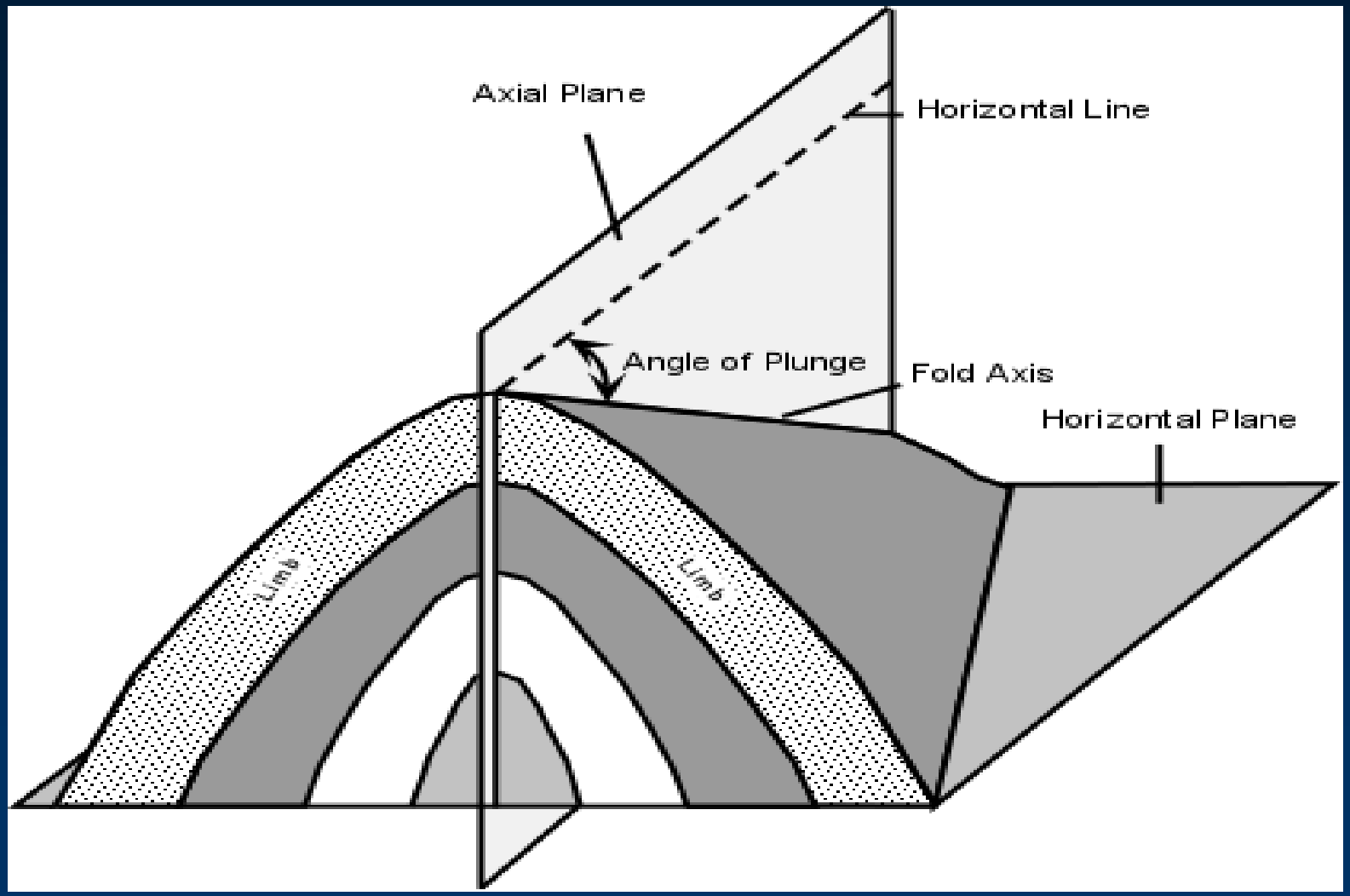
Recall this? It'll help!

Principles of superposition & original horizontality

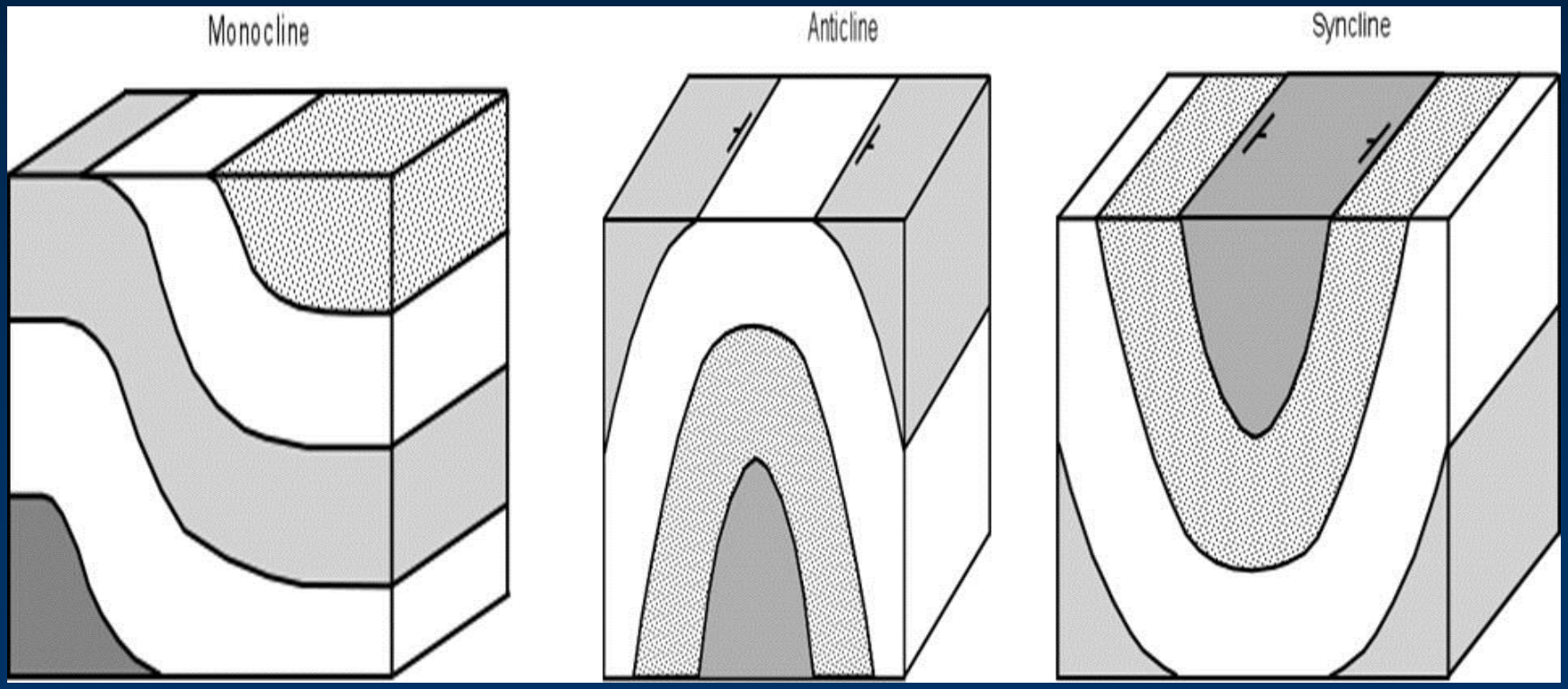


Parts of fold: اجزاء الطية

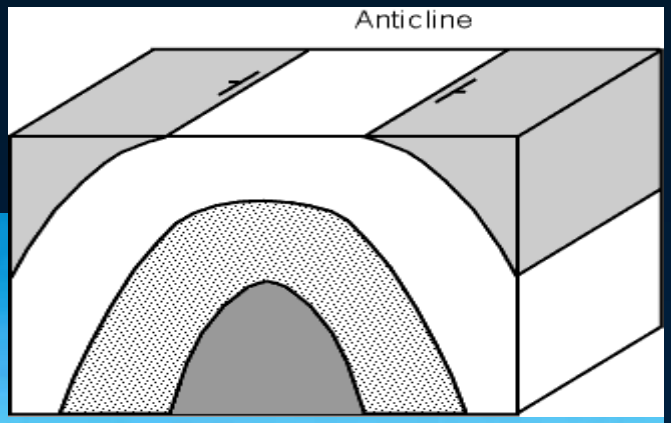
1-Limbs اجنحة 2-Fold axis محور الطية 3-Axial plane مستوى الطية



- Three simple fold forms exist:
  - *Synclines* warp downward طية مقعرة
  - *Anticlines* warp upward طية محدبة
  - *Monoclines* dip in one direction احادى



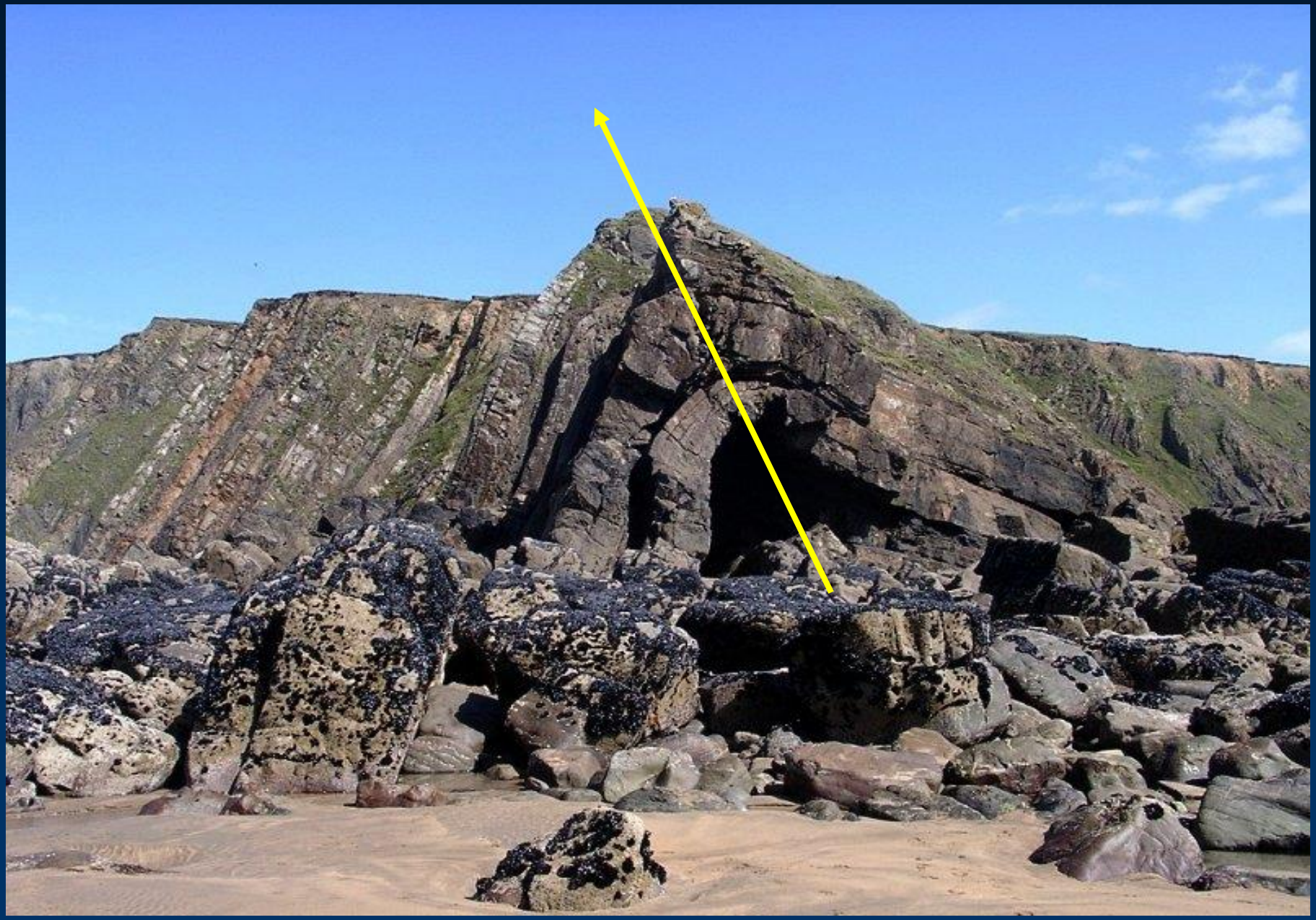
Anticline in its natural environment
-oldest rocks in the core of fold



Syncline: youngest rocks in core of fold

- **Types of Fold**

- Symmetrical Fold: متماثلة two limbs dips in the same
- Asymmetric Fold: غير متماثلة one limb dips more steeply than the other
- Plunging folds: مقلوبة when the folds axis is dipping or plunging
- Overturned and Recumbent: نائمة folding is so extreme that beds are turned upside-down

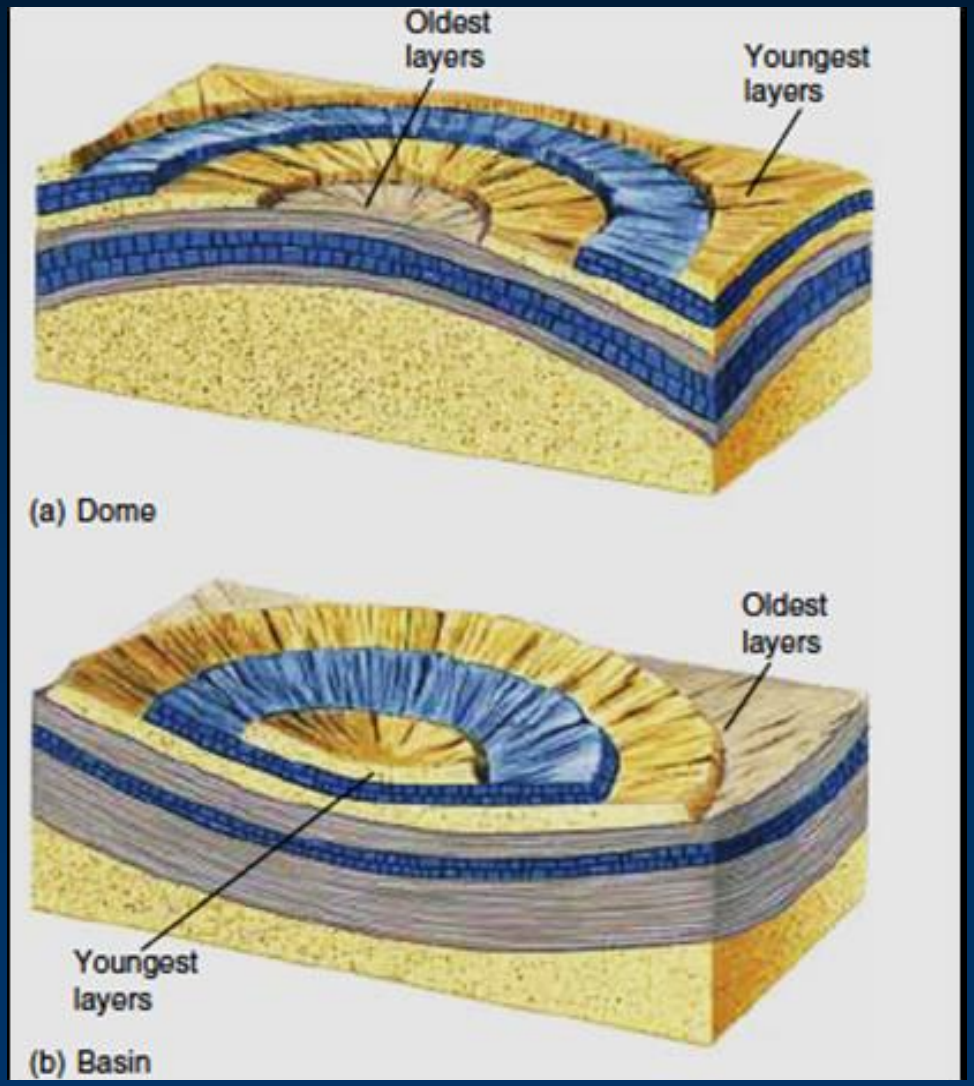


Plunging Fold

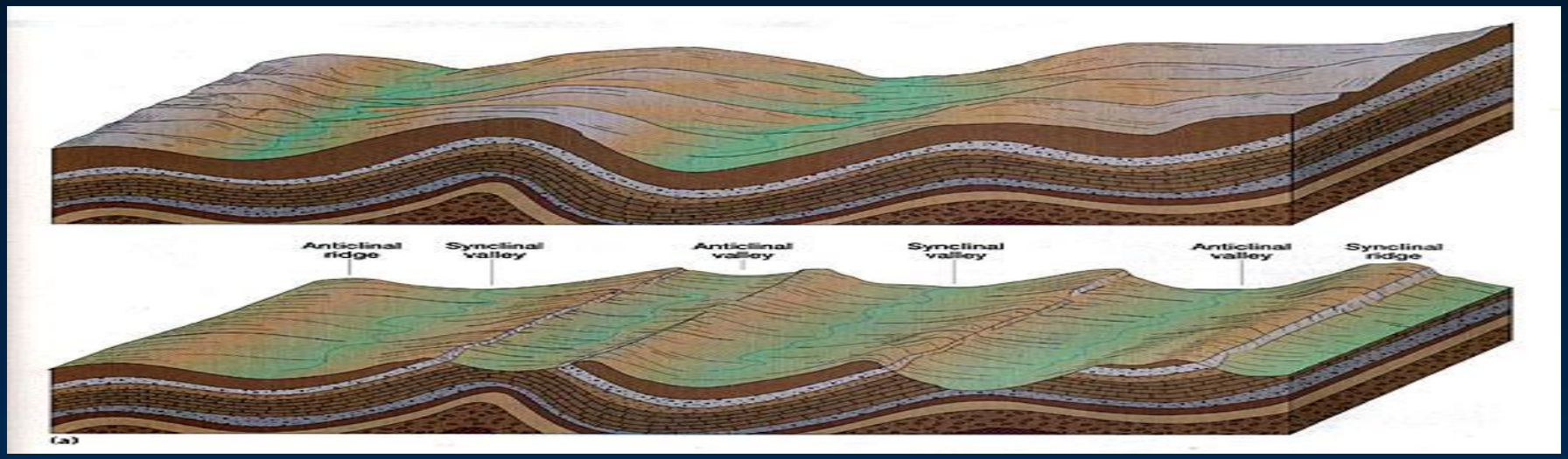
50

Complex Folds

- Domes & Basins حوض- القبة



Folds occur as sets: *“Fold Belts”*



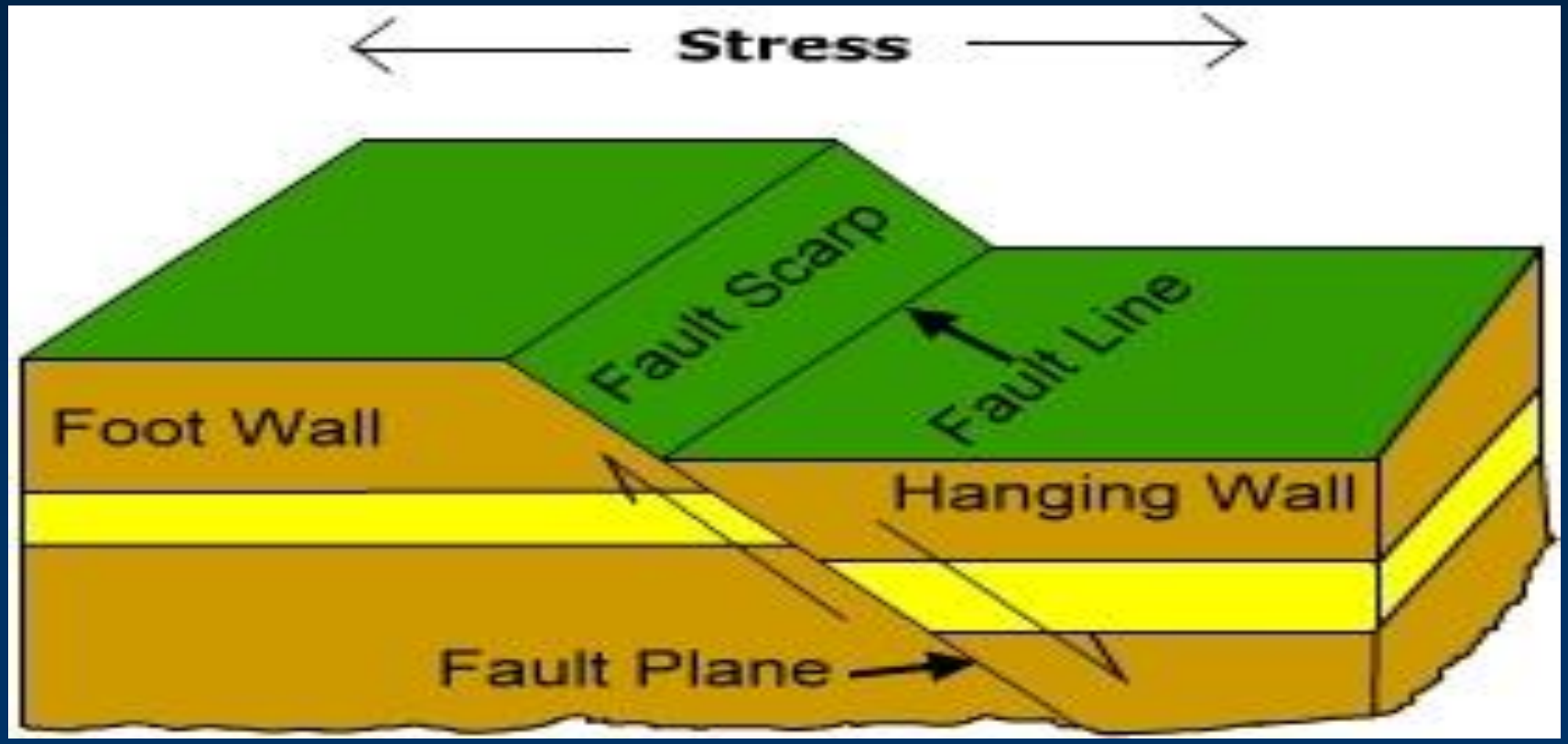
- Folds can be complex
  - Multiple folding events
  - Complex forms are created

Joints

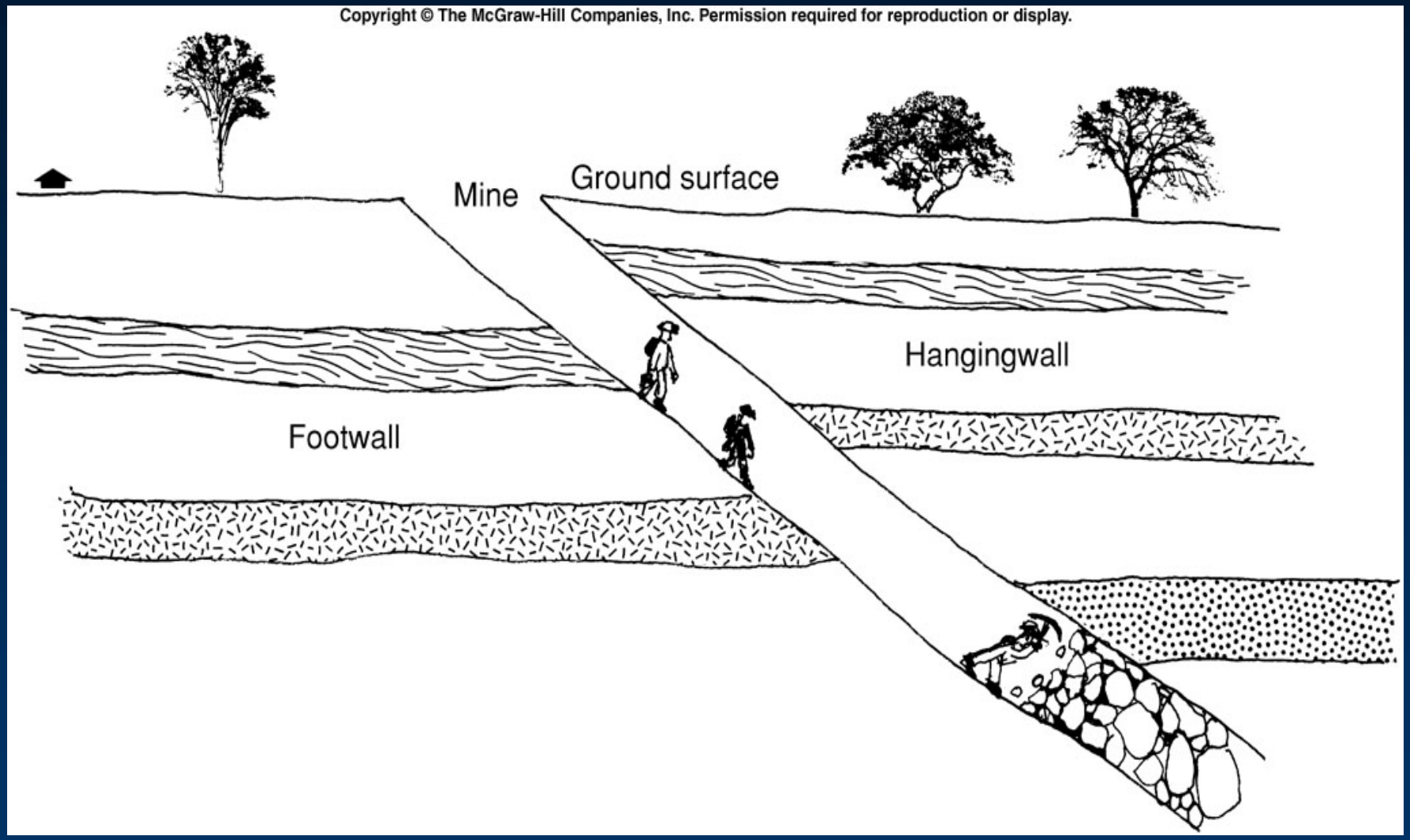
- Fractures created by tension in brittle rocks
  - No shear or displacement has occurred
  - Form as overburden is removed
  - Form by cooling

• Faults

- **Parts of the fault are:**
- 1-Hanging Wall and the Footwall: الحائط العلوى والسفلى
- 2-Strike of the fault: مضرب الفالق
- 3-The dip of the fault: ميل الفالق
- 4-fault plane سطح الفالق Dip ميل Strike مضرب



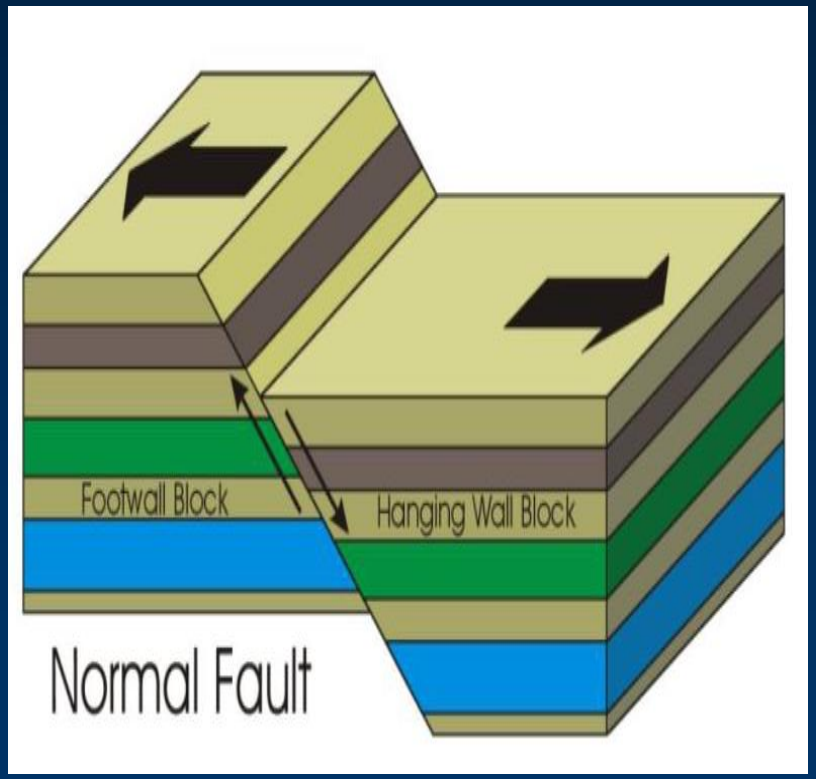
Why "hanging wall" and "footwall?"



Types of faults

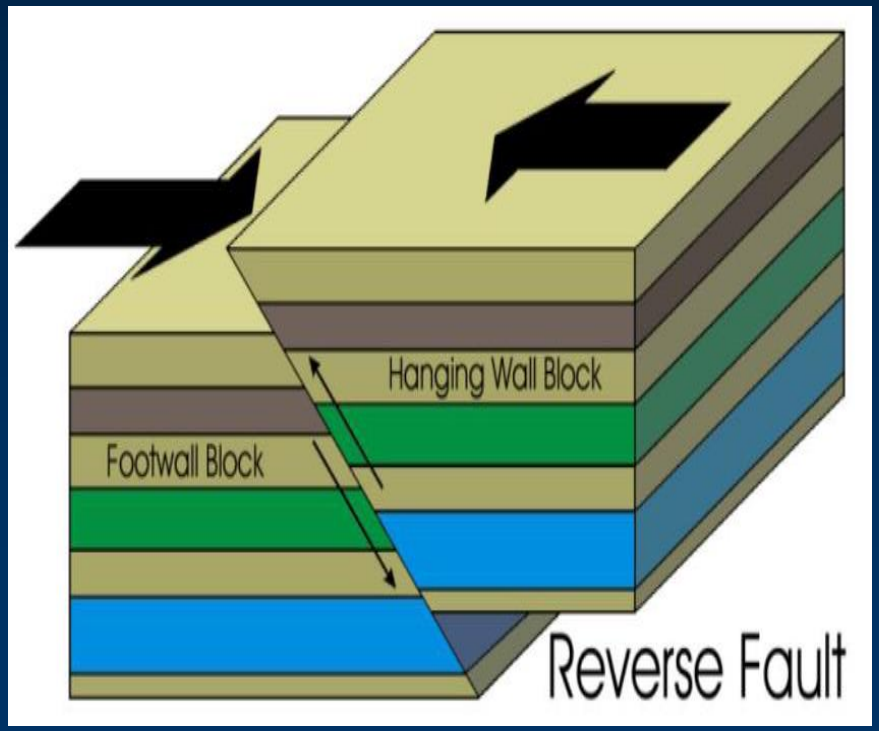
Normal Faults فالق عادى

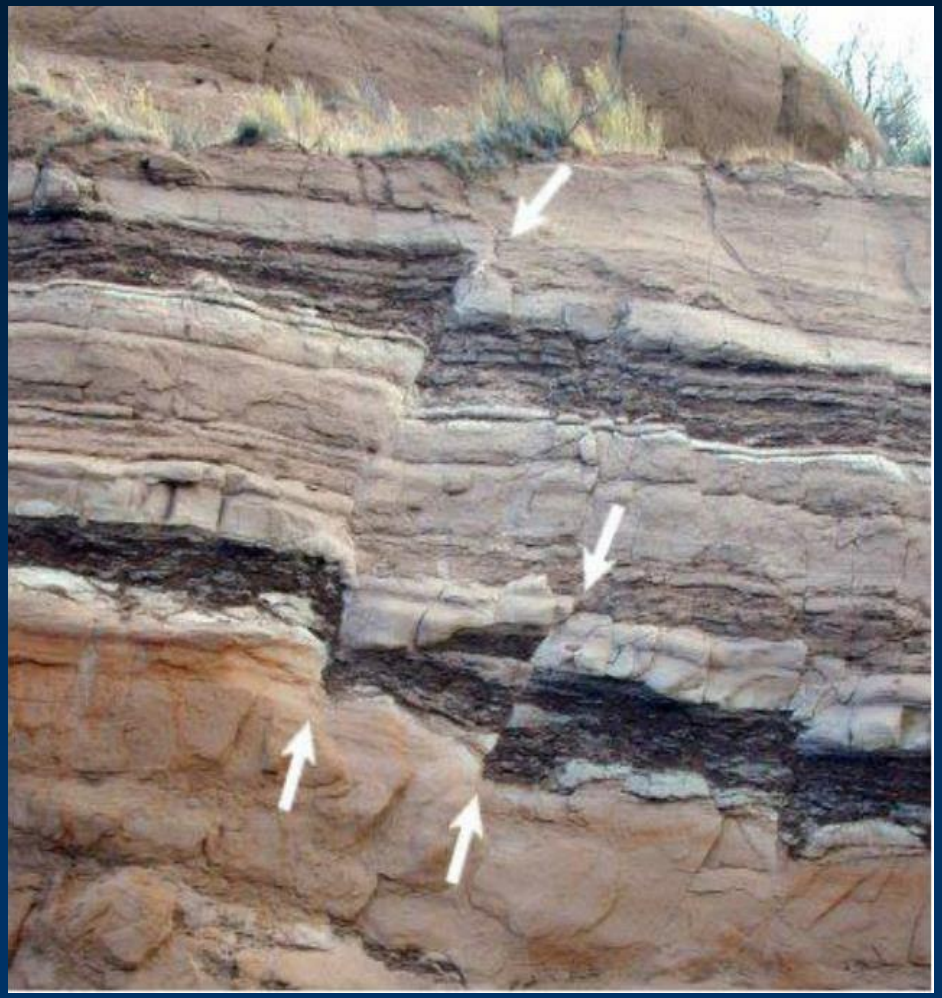
- *Normal faults* are created by tensional forces:



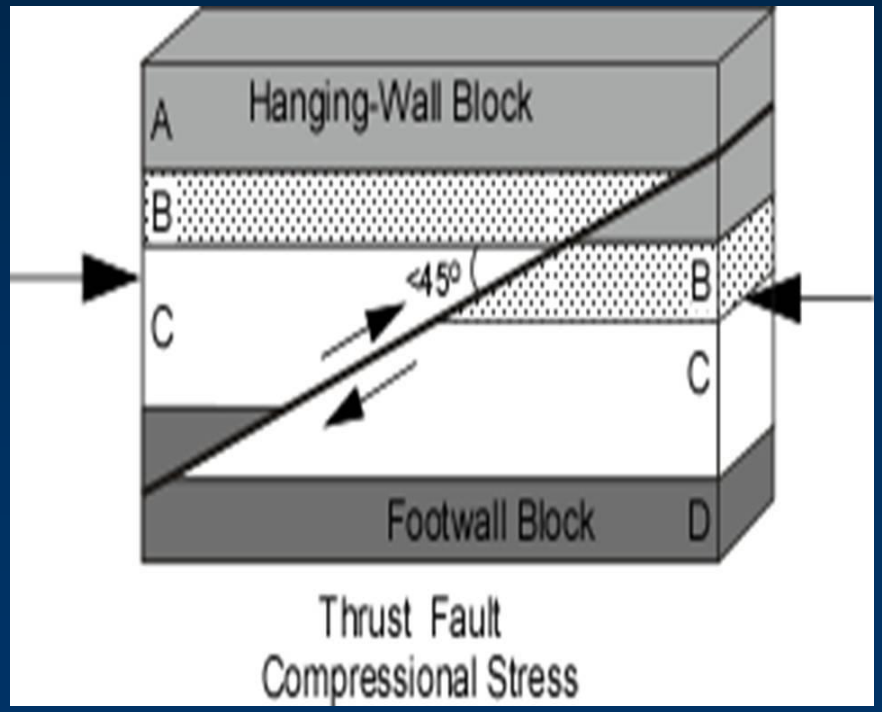
Reverse Faults معكوس

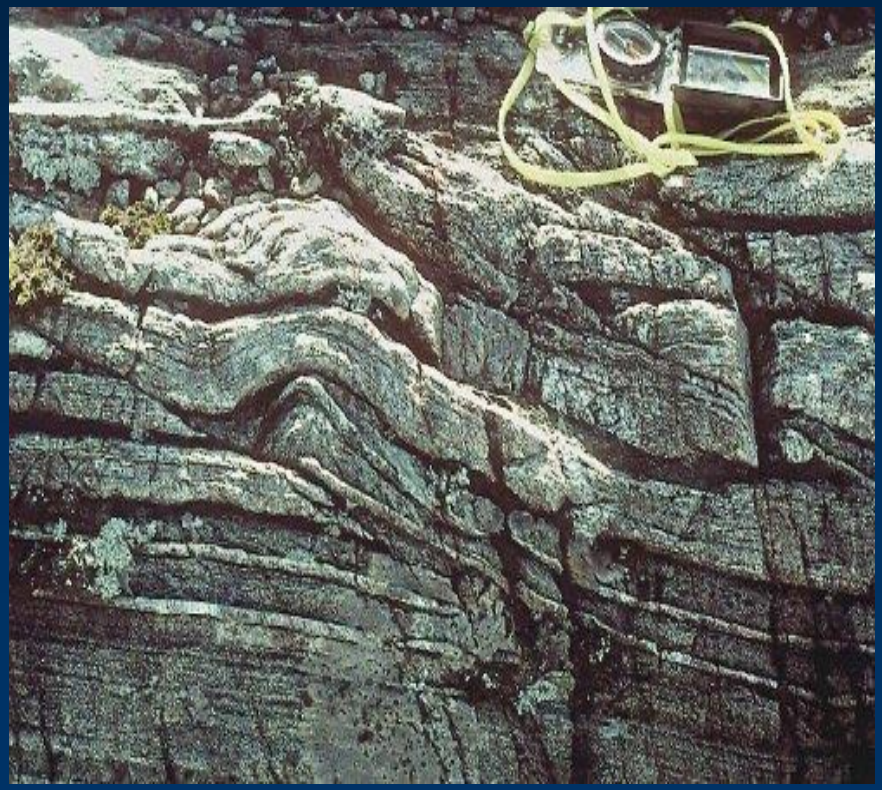
- *Compressional* stress usually causes reverse faults to form





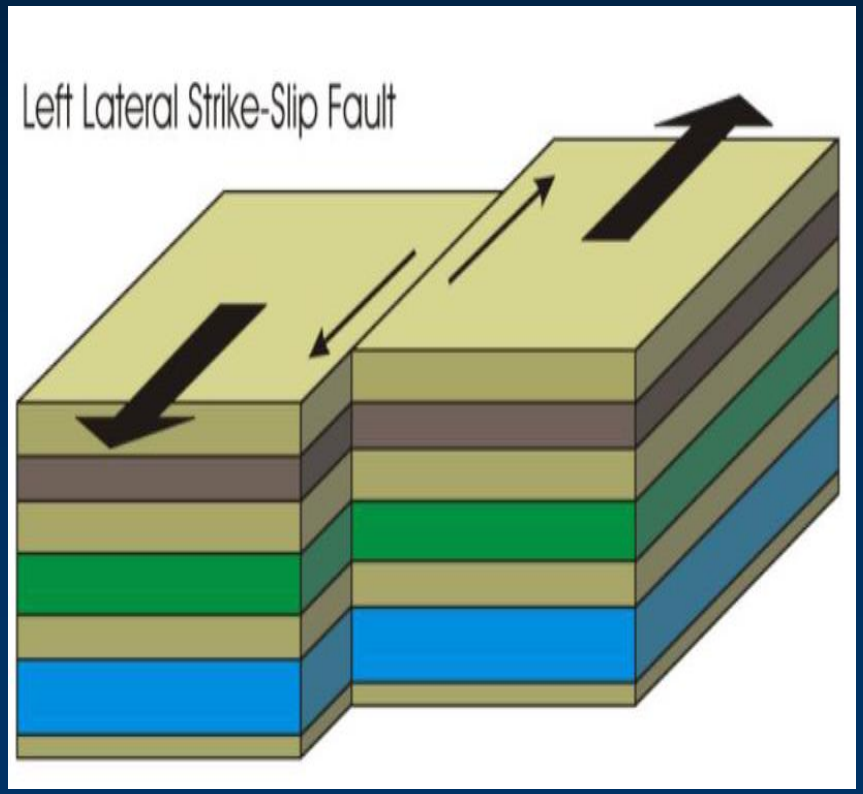
- *Thrust faults* (reverse at low angles)
  - Shallow dip angle, > 45º
  - Horizontal displacement may be tens of kilometers

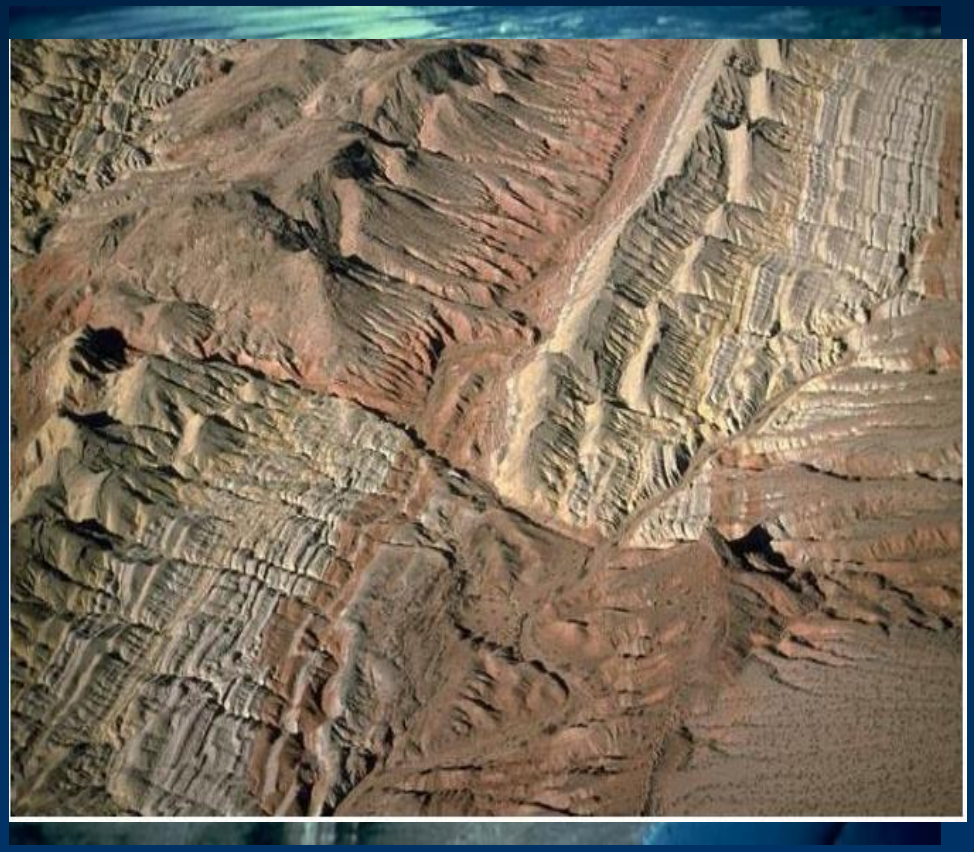


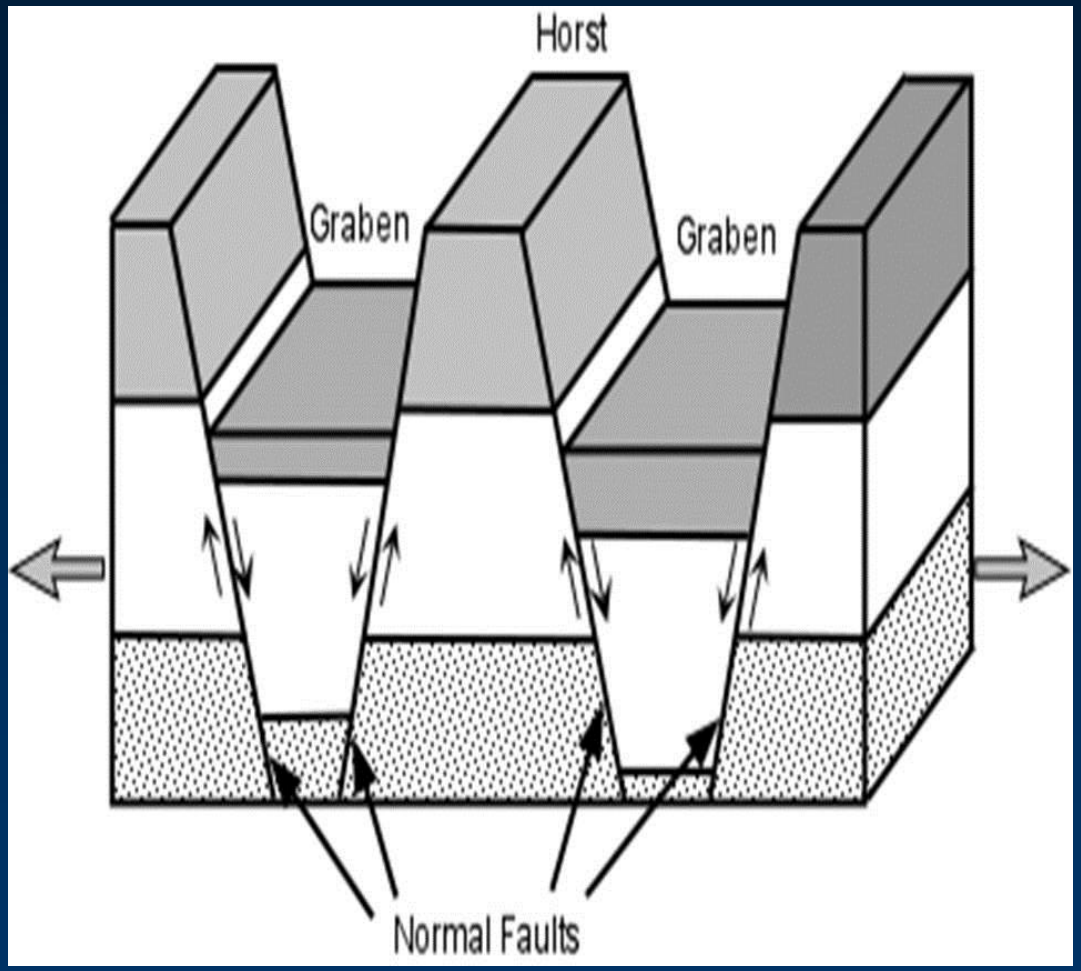


– Strike slip faults فالق المضرب

- Displacement to sideways in a horizontal direction
- Movement is parallel to the strike of the fault plane





Trace of the *San Andreas Fault*, CA

Horst and Graben
هورست اخدود